

Assignment 3

Program -

```
1]
class Stack
{
    constructor()
    {
        this.top = null
    }
    push(ele)
    {
        var node = new newNode(ele)
        node.next = this.top
        this.top = node
    }
    pop()
    {
        var temp = this.top
        var char = temp.data
        this.top = this.top.next
        temp = null
        return char
    }
    reverseString(str)
    {
        var i = 0
        var reversestr = ""
        while (i != str.length)
        {
            this.push(str.charAt(i))
            i++
        }
        var temp = this.top
        while (temp != null)
        {
            var char
            char = this.pop()
            reversestr += char
            temp = this.top
        }
        return reversestr
    }
    display()
    {
        var temp = this.top
        while (temp != null)
        {
            console.log(temp.data)
            temp = temp.next
        }
    }
}
class newNode
{
    constructor(data, next)
```

```
    {
      this.data = data
      this.next = null
    }
  }

const stack = new Stack()
const string = "pratik"
const reverse = stack.reverseString(string)
console.log(`The String provided - ${string}\nString in reverse format - ${reverse}`)
```

Output -

```
[ 1, 2, 4, 8 ]
[ 1, 2, 4 ]
4
false
3
[]
```